

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I Sree B (192524023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B 

19/20
Asst P
29/11/20

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

⑩ Given Network: 192.168.100.0/24

School of Computing (125)

Subnet Mask: 255.255.255.128

Network Address: 192.168.100.0

Broadcast Address: 192.168.100.127

Valid Host Range: 192.168.100.1 - 192.168.100.126

Usable Hosts: $2^7 - 2 = 126$

School of Electronics (126)

Subnet Mask: 255.255.255.192

Network Address: 192.168.100.128

Broadcast Address: 192.168.100.191

Valid Host Range: 192.168.100.129 - 192.168.100.190

Usable Hosts: $2^6 - 2 = 62$

School of Mechanical Engineering (127)

Subnet Mask: 255.255.255.224

Network Address: 192.168.100.192

Broadcast Address: 192.168.100.223

Valid Host Range: 192.168.100.193 - 192.168.100.222

Usable Hosts: $2^5 - 2 = 30$

Remaining Unused IP Range

Unused Block: 192.168.100.224 - 192.168.100.255

Total Remaining Addresses: 32 addresses

SI IPv6 Address

Given Address:

SE NAME

R329

② Given Networks: 10.10.0.0/16

Software Development (124)

Subnet Mask: 255.255.255.0

Network Address: 10.10.0.0

Broadcast Address: 10.10.0.255

Host Range : 10.10.0.1 - 10.10.0.254

Usable Hosts : 254

Quality Assurance (125)

Network Address: 10.10.1.0

Broadcast Address: 10.10.1.127

Host Range : 10.10.1.1 - 10.10.1.126

Usable Hosts : 126

Human Resources (126)

~~Network Address: 10.10.1.128~~

~~Broadcast Address: 10.10.1.191~~

~~Host Range : 10.10.1.129 - 10.10.1.190~~

~~Usable Hosts : 62~~

SIN IPv6 Address

Executive Management (127)

Network Address: 10.10.1.192

Broadcast Address: 10.10.1.223

Host Range : 10.10.1.193 - 10.10.1.222

Usable Hosts : 30

Remaining Unused Range

Unused Address: 10.10.1.224 - 10.10.255.255

③ IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000

1. Compressed Form : 2001:db8:abcd::164

2. Expand Form : 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix. Prefix Length : 164

Network Prefix : 2001:db8:abcd:0000::164

4. Number of Host Address

Host Bits: $128 - 64 = 64$

Hosts: $2^{64} = 18,446,744,073,709,551,614$

5. Total Possible Host Address

18,446,744,073,709,551,616 Addresses

④ Routing Table

Network

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP: 192.168.2.45

1. Destination Subnet:

The IP belongs to 192.168.2.0/24

2. Matching Routing Entry

192.168.2.0/24

3. Next-Hop Interface

Forward the packet through the router.

Interface connected to the 192.168.2.0/24 network

4. Default Route

If no routing entry matches, use

0.0.0.0/0

Default Gateway.

PN Administration LAN (Local Area Network)

FR: 192.168.10.0/24

Frame Local Area Network (LAN): 192.168.10.64/24

Administration Local Area Network (LAN)

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Host Range: 192.168.10.1 - 192.168.10.62

Gateway: 192.168.10.1

Example PC's

* PC1 → 192.168.10.2

* PC2 → 192.168.10.3

* PC3 → 192.168.10.4

Frame Local Area Network

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Host Range: 192.168.10.65 - 192.168.10.126

Gateway: 192.168.10.65

Example PC's

* PC1 $\rightarrow$ 192.168.10.66

* PC2 $\rightarrow$ 192.168.10.67

* PC3 $\rightarrow$ 192.168.10.68

Done
24/11/20